

Midterm

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Set Up

```
#read in packages
library(tidyverse) # for data manipulation like select()
library(ggplot2) # for plots
library(janitor) # for clean_names() in 3d
library(gt) # for table in 3e
library(rstatix) # for Kruskal_test() in 3f

# read in and name data for part 1
halibut <- read_csv("halibut.csv")

# read in and name data for part 2
bee_pathogens <- read_csv("bee-pathogens.csv")
```

Problems

Problem 1. Understanding, critiquing, and adding to written communication

a. Test Type

Two possible test types include a two-sample t-test or a Wilcoxon rank-sum test (also known as the Mann-Whitney U test). Since two groups are being compared in the data set, these tests fit well. Both tests compare the categorical independent variable (North and South) and one continuous dependent variable (halibut landings). Both tests are meant to determine if there is statistical significance between the two groups. Both null hypotheses assume there is no difference between the populations being sampled.

A two-sample t-test is parametric, meaning it assumes data follows normal distribution and specifically assess means. In contrast, the Wilcoxon rank-sum test is non-parametric, which doesn't assume normality. Instead of comparing means it compares the medians or distributions of the two groups using ranks. Non-parametric tests are particularly useful for small data sets or those with skew and outliers, as medians are more robust than means. Statistically the t-test relies on a t-statistic based on means and variances, whereas the Wilcoxon rank-sum test uses a W or U-statistic based on sums of ranks.

b. Writing new statements

Two sample t-test:

Our evidence suggests there is no (Cohen's $d = [0 \text{ or effect size value}]$) difference in mean commercial halibut landings between Northern (mean = [mean], $n = [\text{sample size}]$) and Southern (mean = [mean], $n = [\text{sample size}]$) California from 1969-2019. (Welch's (or Student's) two sample t-test, $t(\text{degrees of freedom}) = \text{test statistic}$, $p = 0.3$, $\alpha = [\text{significance level}]$).

Wilcoxon rank-sum test:

Our evidence suggests there is no (Cliff's $\delta = [0 \text{ or effect size value}]$) difference in distribution or median commercial halibut landings between Northern (median = [median], $n = [\text{sample size}]$) and Southern (median = [median], $n = [\text{sample size}]$) California from 1969-2019. (Wilcoxon rank-sum test, $W = [\text{test statistic}]$, $p = 0.3$, $\alpha = [\text{significance level}]$).

c. New Figures

```
# I am creating a halibut summary to make it easier to plot a confidence interval
# create a new object from clean data frame
halibut_summary <- halibut |>
  # group by location type
  group_by(ca_region) |>
  # calculate values for each production type
  summarize(
    mean = mean(landings_lb), # mean
    n = length(landings_lb), # number of observations (n)
    df = n - 1, # degrees of freedom
    sd = sd(landings_lb), # standard deviation
    se = sd/sqrt(n), # standard error
    tval = qt(p = 0.05/2, df = df, lower.tail = FALSE), # t value
    margin = tval*se, # margin of error
    ci_lower = mean - margin, # lower bound of the confidence interval
    ci_higher = mean + margin) # upper bound of the confidence interval
```

```
# display object
halibut_summary
```

```
# A tibble: 2 x 10
  ca_region    mean     n    df      sd      se  tval margin ci_lower ci_higher
  <chr>      <dbl> <int> <dbl>   <dbl> <dbl> <dbl> <dbl>   <dbl>   <dbl>
1 Northern  390643.   51    50 203163. 28449.   2.01 57141.  333502.  447784.
2 Southern  350715.   51    50 180999. 25345.   2.01 50907.  299808.  401622.
```

```
# base layer: ggplot
ggplot(data = halibut,
       mapping = aes(x = ca_region,
                     y = landings_lb,
                     color = ca_region)) +

# first layer: jitter to display underlying raw data
# custom shape, size and opacity
geom_jitter(width = 0.1,
            height = 0,
            alpha = 0.8,
            shape = 21,
            size = 1) +

# third layer: 95% CI, medium gray, medium width lines show confidence
geom_errorbar(data = halibut_summary,
              mapping = aes(x = ca_region, y = mean,
                            ymin = ci_lower, ymax = ci_higher),
              width = 0.2, color = "gray50") +

# fifth layer: mean point on top shown by large black circle
geom_point(data = halibut_summary, aes(y=mean), color = "black", size = 3)+

# custom colors for jitter points
scale_color_manual(values = c("Northern" = "blue",
                              "Southern" = "magenta")) +

# clean y axis
scale_y_continuous(labels = function (x)x/1000)+

# relabelling x- and y-axes, adding title
labs(x = "California Region",
     y = "Halibut Landings (thousand lbs)",
```

```

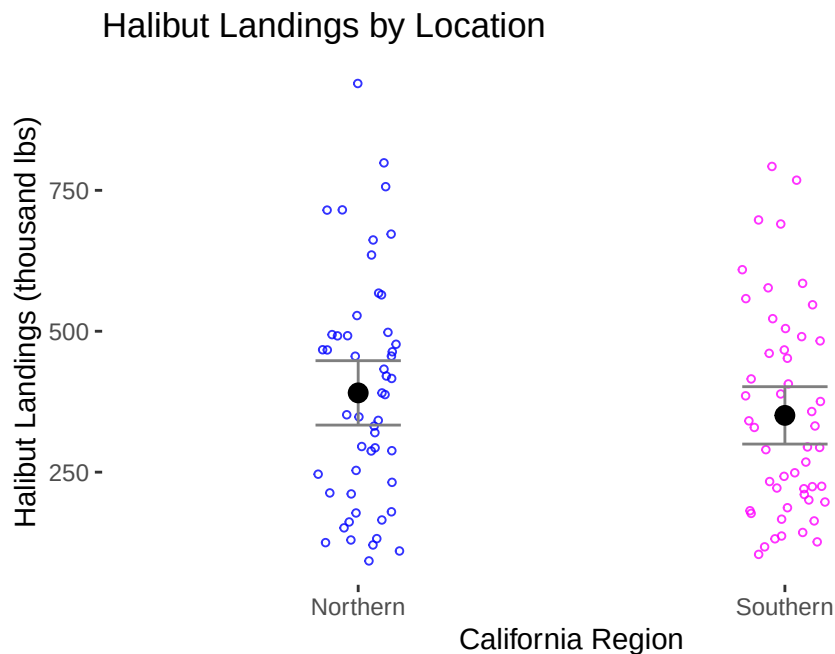
    title = "Halibut Landings by Location") +

# remove legend
  theme(legend.position = "none",

# turn panel background white
    panel.background = element_rect(fill = "white"),

# remove gridlines
    panel.grid = element_blank())

```



```

# base layer
ggplot(data = halibut,
  mapping = aes(x = ca_region,
    y = landings_lb,
    color = ca_region)) +

# Layer 1: boxplot
# Boxplots show median with thick horizontal line, I made it thicker for
# visual clarity. Boxplots show spread with IQR box and whiskers.
# The length of the IQR box displays precision (shorter = more precise)
# Apparently notches are the way to display 95% confidence so I added that in
# light blue

```

```

geom_boxplot(median.linewidth = 1.5, notch = TRUE, fill = "lightblue")+

# Layer 2: jitterplot for underlying data points
# Height is 0 to ensure no vertical spread, custom width, size, shape,
# & transparency
geom_jitter(height = 0,
            width = 0.1,
            shape = 20,
            alpha = 0.5,
            size = 2)+

# Labeling axes
labs(y = "Halibut Landings (thousand lbs)", x = "California Region")+

# change colors form default
scale_color_manual(values = c("blue3","magenta3"))+

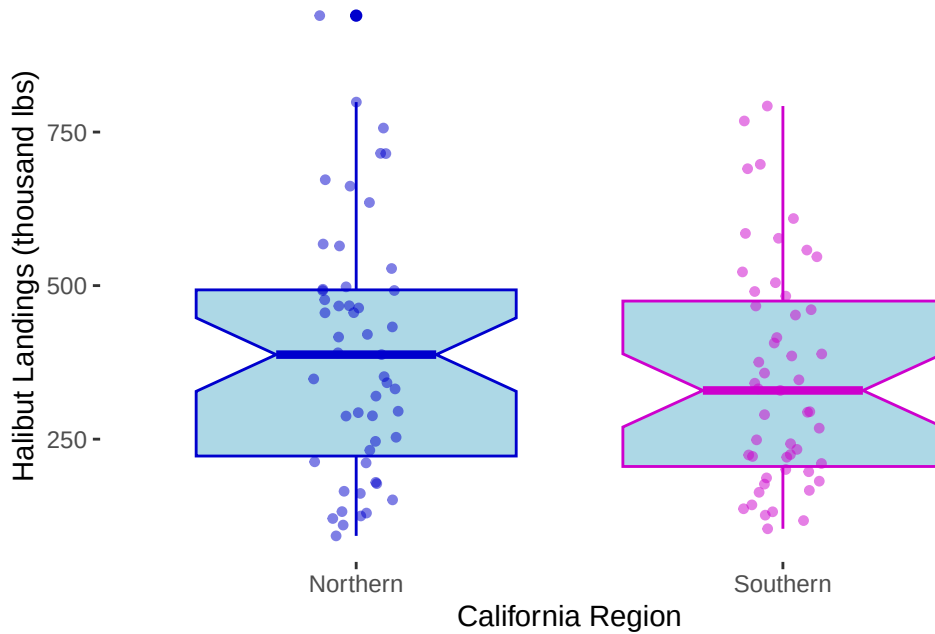
# clean y axis
scale_y_continuous(labels = function (x)x/1000)+

# remove legend
theme(legend.position = "none",

# turn panel background white
      panel.background = element_rect(fill = "white"),

# remove gridlines and
      panel.grid = element_blank())

```



d. Caption writing

Figure 1. The mean halibut landings are not significantly different between Northern and Southern California The two regions being observed are represented on the x-axis and the pounds of halibut landing annually are represented on the y axis. The annual mean halibut landings are represented by large black, opaque, circles. The hollow blue and magenta circles represent data points from Northern and Southern California respectively. The gray bar represents a 95% confidence interval. Data originally from: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Figure 2. The median halibut landings between Northern and Southern California are not significantly different. The two regions being observed are represented on the x-axis and the pounds of halibut landing annually are represented on the y axis. The annual median halibut landings are represented by thick horizontal lines. The semi-transparent blue and magenta circles represent data points from Northern and Southern California respectively. The solid blue circle is a potential outlier. The light blue box represents the Interquartile range. The notches display a 95% confidence interval. Data originally from: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 2. Data Visualization

a. Plan your figure

x-axis: “year”

y-axis: “landings_lb”

geometry function: `geom_line()`

colors and linetypes: column: `ca_region`, Northern: dotted blue line, Southern: solid orange line.

relabelling the x- and y-axes: `labs(x=“YEAR”,y=“HALIBUT LANDINGS (LB)”)`

using custom colors: `scale_color_manual()`

theme arguments and the element for: all of the following in `theme()`

positioning the x-axis text at a 45° angle: `axis.text.x = element_text(angle = 45, hjust = 1)`

putting the legend in the top right of the plot: `legend.position = “inside”, legend.justification = c(“right”, “top”)`

removing the panel background: `panel.background = element_blank()` to remove or `panel.background = element_rect(fill = “white”)` to turn it white

removing the panel grid: `panel.grid = element_blank()`

removing the axis ticks: `axis.ticks = element_blank()`

removing the legend title: `legend.title = element_blank()`

b. Code your figure

```
# remove scientific notation globally to match given graph
options(scipen = 999)

# base layer, assign x and y values, and separates lines by color and type
ggplot(data = halibut,
       mapping = aes(x = year,
                     y = landings_lb,
                     color = ca_region,
                     linetype = ca_region))+
  geom_line(linewidth = 0.9)+ #this makes the line a little thicker

# this shows more years on the x axis than the default (decades)
```

```

scale_x_continuous(breaks = seq(min(halibut$year),
                                max(halibut$year), by = 2))+

# this sets the y axis range and increments to match the given image
scale_y_continuous(limits = c(0,1000000),
                   breaks = seq(0, 1000000, by = 100000),
                   expand = expansion(mult = c(0, 0.05)))+

# Assigns specific colors to lines/regions
# changed legend to be capitalized text
scale_color_manual(values = c("Northern" = "#4472C4","Southern" = "#ED7D31"),
                   labels = c("Northern" = "NORTHERN CALIFORNIA",
                              "Southern" = "SOUTHERN CALIFORNIA"))+

# Assigns line type to regions and changed legend to be capitalized text
scale_linetype_manual(values = c("Northern" = "dotted","Southern" = "solid"),
                      labels = c("Northern" = "NORTHERN CALIFORNIA",
                                 "Southern" = "SOUTHERN CALIFORNIA"))+

# label x and y axis like given graph, add name as subtitle
labs(x="YEAR",
     y="HALIBUT\nLANDINGS (LB)",
     subtitle = "Jocelyn Magner")+

# adjust theme to match given graph
theme(

  # reduce text size of axis text and legend text to reduce overlap
  axis.title = element_text(size = 8),
  axis.text = element_text(size = 6),
  legend.text = element_text(size = 6),

  # rotate axis title
  axis.title.y = element_text(angle = 0, vjust = 0.5),

  # rotate years 45 degrees
  axis.text.x = element_text(angle = 45, vjust = 0.5),

  # add back gray x axis
  axis.line.x = element_line(color = "#D9D9D9"),

  # move legend
  legend.position = "inside", legend.justification = c("right", "top"),

```

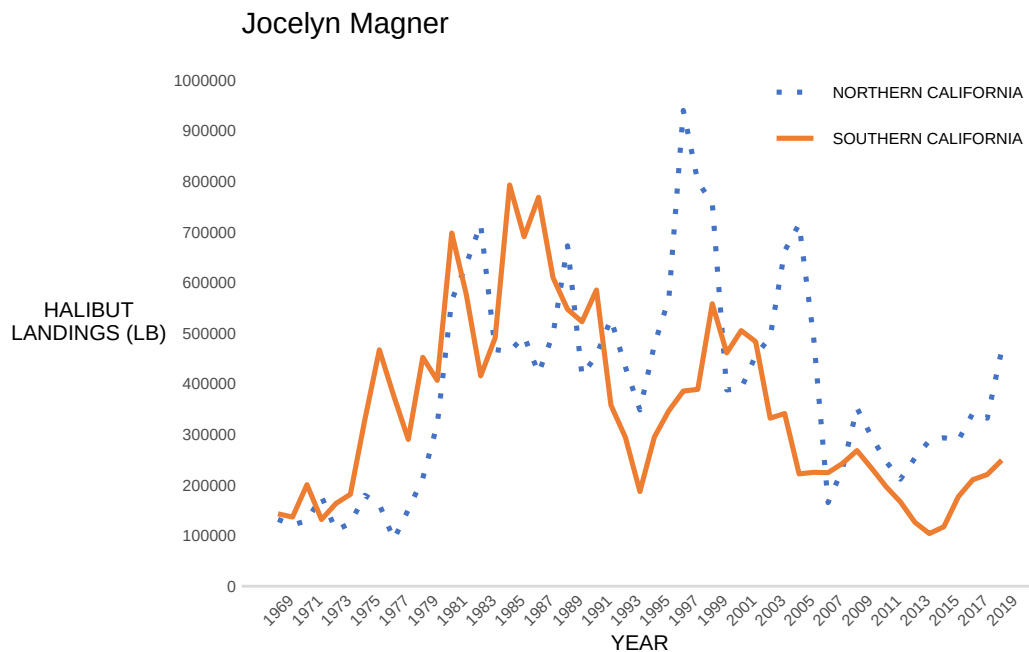


```
# remove background
panel.background = element_blank(),

# remove grid
panel.grid = element_blank(),

# remove axis ticks
axis.ticks = element_blank(),

# remove legend title
legend.title = element_blank())
```



c. Write a caption

Figure 3. Halibut landings (in lbs) in Northern and Southern California trends vary from 1969 to 2019. Southern California is represented by the solid orange line and Northern California is represented by the dotted blue line. The landing weights in Northern and Southern California follow a similar trend from 1969 to 1993, having peaks around the early 1980s and declining around 1993. From 1993 to 2019 it appears that Northern California tends to have consistently higher landing weights compared to Southern California. Data originally from: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 3. Reproducing an analysis

Components

a. Variables

The response variable analyzed in this sentence is Black Queen Cell Virus load. It is a continuous variable. The researchers used random sampling, sample pooling, RNA extraction, reverse transcription, and quantitative PCR tests to measure BQCV load. They compared this to a control variable, so rather than having units, the loads are quantified as log2 fold change. This information is found in the Methods section, under Pathogen Quantification (page 493).

The predictor variable is habitat type (Forest, Edge, and Valley). This information is found many places, but can also be seen in page 493 in figure 1.

b. Test Type

Kruskal-Wallis test, Dunn's Post-hoc, and Bonferroni correction. The test type is discussed in the Statistical Analysis section (the sixth section, the 11th paragraph) of the Methods section.

c. Hypothesis

Null hypothesis: The population medians of Black Queen Cell Virus load is the same across all habitat types.

Alternative hypothesis: At least one habitat type differs in its population median of Black Queen Cell Virus load.

d. Cleaning and organizing data

```
#creating new object
bee_pathogens_clean <- bee_pathogens|>

# clean column names
clean_names()|>

# select only columns of interest
select(site_type_x, bqcv)|>
```

```
# rename site column to habitat
rename(habitat = site_type_x)

# Display 10 rows from cleaned data set
bee_pathogens_clean |> slice_sample(n = 10)
```

```
# A tibble: 10 x 2
  habitat    bqcw
  <chr>    <dbl>
1 forest    0
2 valley  4.02
3 forest  1.000
4 valley  8.42
5 forest  1.26
6 valley  8.45
7 forest  5.56
8 valley  9.42
9 valley  4.58
10 forest  3.98
```

e. Summarizing and table displays

```
# create new object
bee_pathogens_summary <- bee_pathogens_clean |>

# separate by habitat
group_by(habitat) |>

# find and round median bqcw, ignore any missing values, add sample size column
summarize(median_bqcw = round(median(bqcw, na.rm = TRUE), 1),
          sample_size = n())
```

```
#creating table
bee_pathogens_summary |>
gt() |>

# add titles
tab_header(title = "BQCV Load by Habitat",
           subtitle = "Median values and sample sizes") |>
```

BQCV Load by Habitat

Median values and sample sizes

Habitat Type	Median BQCV	n
edge	6.3	31
forest	5.0	26
valley	7.6	30

```
# polish column names
cols_label(habitat = "Habitat Type",
           median_bqcv = "Median BQCV",
           sample_size = "n")|>

# align column left for visual clarity
cols_align(align = "left")
```

f. Do a test

```
# They used a Kruskal-Wallis test. Use cleaned data
kruskal.test(bqcv ~ habitat,
             data = bee_pathogens_clean)
```

Kruskal-Wallis rank sum test

data: bqcv by habitat

Kruskal-Wallis chi-squared = 9.8219, df = 2, p-value = 0.007365

g. Do a post-hoc test

```
# Use Dunn's post-hoc test, use clean data
dunns <- dunn_test(bqcv ~ habitat,
                  data = bee_pathogens_clean,
                  p.adjust.method = "bonferroni")
```

h. Calculate effect size

```
# Calculate effect size with cleaned data
kruskal <- kruskal_effsize(bqcv ~ habitat,
                           data = bee_pathogens_clean)
```

i. Written communication

There was a **moderate** effect ($\eta^2 = 0.09$) of habitat type on BQCV loads (**Kruskal-Wallis test**, $\chi^2(2) = 9.82$, $P = 0.007$, $\alpha = 0.05$; **Table 1**). **Pairwise comparisons using Dunn's Post-hoc test with Bonferroni adjustment revealed** significant differences between valley (**median = 7.6, n = 29**) and forest (**median = 5.0, n = 24**; $z = 3.13$, **P-adj = 0.005**), but edge (**median = 6.3, n = 27**) was not different from forest ($z = -1.82$, **P-adj = 0.203**) nor valley ($z = 1.31$, **P-adj = 0.567**).